JIE GAO

Name: Jie Gao **Phone:** +86 18822141866 **Email:** jiegao@tjnu.edu.cn
Date of Birth: 22/02/1996 **ORCID:** https://orcid.org/0000-0002-3572-6966



WORK EXPERIENCE

Associate Research Fellow
Academy of Eco-civilization Development for Jing-Jin-Ji Megalopolis, Tianjin Normal University, China 08/2025–Present

EDUCATION

Ph.D., Doctor of Science in Environmental Science 09/2021–06/2025
Nankai University, China. Supervisor: Prof. Guoliang Shi, Prof. Shaojie Song

Visiting Doctoral Researcher (China Scholarship Council) 02/2024–08/2024
University of Birmingham, United Kingdom. Supervisor: Prof. Zongbo Shi

M.S., Master of Science in Environmental Science 09/2018–06/2021
Nankai University, China. Supervisor: Prof. Guoliang Shi, Prof. Yinchang Feng

B.E., Bachelor of Engineering in Environmental Ecological Engineering 09/2014–06/2018
Shanxi University, China.

SELECTED HONORS AND AWARDS

- First Prize, the Tang Xiaoyan Environmental Science Innovation Award, Peking University, China (5 winners nationwide), 05/2025.
- Nomination Award, the Outstanding Young Doctoral Student Scholarship, Gao Tingyao Environmental Protection Science and Technology Development Foundation, Tongji University, China (10 nominees nationwide), 08/2024.
- National First Prize and Outstanding Supervisor Award, Environmental Engineering Simulation Innovation Track, 4th National Simulation Innovation Application Competition, Talent Exchange Center of the Ministry of Industry and Information Technology, China, 11/2025.
- National Scholarship (twice), Ministry of Education of the People's Republic of China, 12/2022 and 12/2020.
- Tianjin Municipal Outstanding Student, China, 02/2023.
- Outstanding Master's Thesis Award, Tianjin Degree and Graduate Education Association, China, 04/2023.
- Outstanding Doctoral Graduate and Outstanding Master's Graduate, Nankai University, China, 04/2025 and 04/2021.

RESEARCH INTERESTS

Air Pollution Control, Source Apportionment of Atmospheric Particulate Matter, Secondary Aerosol Formation Mechanism, Atmospheric Chemical Modeling, Data Analysis, Environmental Factors.

PUBLICATIONS

Total publications: 20 papers (11 papers as first/corresponding author)

1. **Gao, J.** †, Wang, H. †, Liu, W., Xu, H., Wei, Y., Tian, X., Feng, Y., Song, S.*, Shi, G.*. Hydrogen peroxide serves as pivotal fountainhead for aerosol aqueous sulfate formation from a global perspective. ***Nature Communications***, 2024, 15, 4625. https://doi.org/10.1038/s41467-024-48793-1. (JCR Q1, IF=18.1)
2. **Gao, J.**, Shi, G.*, Zhang, Z., Wei, Y., Tian, X., Feng, Y., Russell, A. G., Nenes, A. Targeting atmospheric oxidants can better reduce sulfate aerosol in China: H_2O_2 aqueous oxidation pathway dominates sulfate formation in haze. ***Environmental Science & Technology***, 2022, 56, (15), 10608-10618. https://doi.org/10.1021/acs.est.2c01739. (JCR Q1, IF=12.2, ES&T Supplementary Cover Article)
3. **Gao, J.**, Wei, Y., Wang, H., Song, S., Xu, H., Feng, Y., Shi, G.*, Russell, A. G. Multiphase buffering: A mechanistic regulator of aerosol sulfate formation and its dominant pathways. ***Environmental Science & Technology***, 2025, 59(16), 8073-8084. https://doi.org/10.1021/acs.est.4c13744. (JCR Q1, IF=12.2)
4. **Gao, J.**, Li, Y., Li, J., Shi, G.*, Liu, Z.*, Han, B., Tian, X., Wang, Y., Feng, Y., Russell, A. G. Impact of formation pathways on secondary inorganic aerosol during haze pollution in Beijing: Quantitative evidence from high-resolution observation and modeling. ***Geophysical Research Letters***, 2021, 48, e2021GL095623. https://doi.org/10.1029/2021GL095623. (JCR Q1, IF=5.0)

5. Yang, Q., Zhao, D., Ma, T., **Gao, J.***, Xu, H., Feng, Y., Shi, G.*. Development and diagnosis of a temporal-context U-Net deep learning surrogate model for CMAQ particulate nitrate over China. ***Environmental Science & Technology Letters***. https://doi.org/10.1021/acs.estlett.6c00755. (JCR Q1, IF=7.3)
6. Li, J., Xu, H., Gong, J., **Gao, J.***, Song, S.*, Sun, Q., Feng, Y., Shi, G. Role of $PM_{2.5}$ sources and environmental drivers in the triangular relationship of nitrate-sulfate-ozone system. ***Journal of Cleaner Production***, 2026, 563, 148516. https://doi.org/10.1016/j.jclepro.2026.148516. (JCR Q1, IF=10.7)
7. **Gao, J.**, Wei, Y., Zhao, H., Liang, D., Feng, Y., Shi, G.*. The role of source emissions in sulfate formation pathways based on chemical thermodynamics and kinetics model. ***Science of the Total Environment***, 2022, 851, 158104. https://doi.org/10.1016/j.scitotenv.2022.158104.
8. **Gao, J.**, Dong, S., Yu, H., Peng, X., Wang, W., Shi, G.*, Han, B., Wei, Y., Feng, Y. Source apportionment for online dataset at a megacity in China using a new PTT-PMF model. ***Atmospheric Environment***, 2020, 229, 117457. https://doi.org/10.1016/j.atmosenv.2020.117457. (JCR Q2, IF=3.8)
9. **Gao, J.**, Wei, Y., Shi, G.*, Yu, H., Zhang, Z., Song, S., Wang, W., Liang, D., Feng, Y. Roles of RH, aerosol pH and sources in concentrations of secondary inorganic aerosols, during different pollution periods. ***Atmospheric Environment***, 2020, 241, 117770. https://doi.org/10.1016/j.atmosenv.2020.117770. (JCR Q2, IF=3.8)
10. Yang, Q., **Gao, J.***, Chen, J., Ma, T., Xu, H., Wang, F., Wei, Y., Liu, J., Sun, Q., Feng, Y., Shi, G. Interpretable machine learning for exploring multiple influencing factors of $PM_{2.5}$ and their interaction effects. ***Journal of Environmental Sciences***, 2026, 105560. https://doi.org/10.1016/j.jes.2026.105560. (JCR Q1, IF=6.5)
11. **Gao, J.**, Shi, X., Wei, Y., Song, S., Shi, G.*, Feng, Y. Evaluation of different ISORROPIA-II modes and the influencing factors of aerosol pH based on Tianjin online data. ***Environmental Science***, 2020, 41(8), 3458–3466. https://doi.org/10.13227/j.hjkx.201912221. (In Chinese)

In process

12. **Gao, J.**, Zhao, D., Ma, T., Yang, Q., Xu, H., Song, S., Feng, Y., Shi, G.*. ***The Innovation Geoscience***. (under review).

RESEARCH PROJECTS

Principal Investigator (PI)

- **National Natural Science Foundation of China (NSFC) Young Scientists Fund (C)**. 01/2027–12/2029
  Study on the influence of emission sources on secondary inorganic aerosol formation under the acid-base buffering effect of aerosols, CNY 300,000.
- **Open Fund of the State Environmental Protection Key Laboratory of Urban Air Particulate Matter Pollution Prevention and Control.** 01/2026–12/2027
  Dominant oxidation pathways and key driving mechanisms of sulfate during pollution periods, CNY 20,000.
- **Tianjin Natural Science Foundation for Young Scientists (B), under review.** 10/2026–09/2028
  Study on competition-synergy mechanism of key components in secondary inorganic aerosols and fine source apportionment of priority control sources, CNY 60,000.

Core Team Member

- **National Natural Science Foundation of China (NSFC) General Program.** 01/2018–12/2021
  Aerosol acidity characteristics and their interactions with emission sources under complex air pollution conditions.
- **National Natural Science Foundation of China (NSFC) General Program.** 01/2021–12/2024
  Understanding the key factors affecting the growth of secondary inorganic aerosol during different pollution processes using online observed data and hybrid model.
- **National Key Research and Development Program of China.** 12/2023–11/2026
  Three-Dimensional Observation, Transport Mechanisms, and Formation Processes of Complex Air Pollution in Mountainous Regions.
- **National Environmental Standard Development Project.** 02/2024 released
  Ambient air—Source apportionment on particulate matter—Technical specification for monitoring data processing and inspection with receptor models based on manual monitoring.

PATENT AND SOFTWARE

- Shi, G., **Gao, J.**, Feng, Y. Secondary sulfate formation rate calculation software via aqueous-phase pathways V1.0. Registered Software Copyright, National Copyright Administration of China, 2023. Registration No.: 2023SR487389.
- Shi, G., Wei, Y., **Gao, J**. Calculation software for gas-phase pathway formation rate of secondary nitrate V1.0. Registered Software Copyright, National Copyright Administration of China, 2021. Registration No.: 2022SR0220103.
- Shi, G., Gong, J., **Gao, J.**, Zhao, D., Feng, Y. Calculation software for the formation rate of atmospheric secondary sulfate through different aqueous-phase oxidation pathways. Registered Software Copyright, National Copyright Administration of China, 2025. Registration No.: 2025SR1378353.
- Shi, G., Zhao, D., Xu, B., **Gao, J.**, Feng, Y. Positive definite matrix factorization regularized supervised scientific computing software 2.0. Registered Software Copyright, National Copyright Administration of China, 2025. Registration No.: 2025SR2164126.
- Shi, G., Zhang, Z., Zhao, D., **Gao, J.**, Wang, F., Feng, Y. Emission Calculation and Scenario Analysis Software Based on Energy Data. Registered Software Copyright, National Copyright Administration of China, 2025. Registration No.: 2025SR1378373.

TECHNICAL EXPERTISE

- Machine learning and scientific programming in Python, chemical thermodynamics and kinetic calculation.
- Atmospheric modeling: GEOS-Chem, WRF-CMAQ, ISORROPIA-II, E-AIM, PMF, ME2, Meteoinfo.

PROFESSIONAL SERVICE AND POLICY SUPPORT

- Trainer for environmental management agency, the Beijing Municipal Ecological and Environmental Monitoring Center.
- Contributed to the Ministry of Ecology and Environment's air pollution control program and received official certification.

ACADEMIC SERVICES

- Reviewer for *Atmospheric Chemistry and Physics, Environmental Pollution, Atmospheric Environment, Urban Climate, Environmental Toxicology and Chemistry, Environmental Chemistry.*
- Topic Editor, Formation Mechanisms and Multi-Scale Control Strategies for Atmospheric Composite Pollution, Participating Journals including *Atmosphere, Processes, Remote Sensing, Standards, Toxics.*

CONFERENCE PRESENTATIONS

10^+ presentations at national and international conferences in atmospheric science and environmental research.

- Invited Oral Presentation, 2nd Future Atmospheric Science Forum, Suzhou, China, 04/2026.
- Oral Presentation and Session Chair, 31st Conference on Atmospheric Environmental Science and Technology and Symposium on Air Pollution Prevention and Control Technology, Tianjin, China, 10/2025.
- Outstanding Oral Presentation, 1st Peking University Atmospheric Environmental Chemistry Doctoral Symposium, Online, 08/2022.
- Oral Presentation (Grand Prize), 11th Nankai University–Tianjin University Doctoral Forum on Ecology and Environment, Tianjin, China, 11/2022.
- Oral Presentation (Grand Prize), 1st Tianjin Graduate Forum on Environmental Science and Engineering, Tianjin, China, 05/2025.
- Oral Presentation, Atmospheric Group Seminar, University of Birmingham, Birmingham, United Kingdom, 06/2024.
- Oral Presentation, 27th and 28th Conference on Atmospheric Environmental Science and Technology, online, 11/2021 and 11/2022.
- Oral Presentation, 8th Symposium on Cost-Benefit Analysis and Attainment Assessment of Air Pollution Control, Guangzhou, China, 11/2020.
- Oral Presentation, 14th and 16th National Conference for Environmental Doctoral Students, online, 10/2020 and 10/2022.
- Poster Presentation, 2nd World Universities Climate Change Alliance Graduate Forum and Tsinghua University International Graduate Forum, Beijing, China, 11/2020.

CONFERENCE ORGANIZATIONS

- Lead Academic Volunteer, International Workshop on Aerosol Acidity and Secondary Particles, Tianjin, China, 12/2019.
- Conference Volunteer, 10th National Conference on Environmental Chemistry, Tianjin, China, 08/2019.